The Reach of K and S: A Program Note on Demonstrations in Mathematics and Physics

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Abstract

Papers 1 and 2 built a number system and a transparent arithmetic inside the two-combinator calculus {K, S}, with the strongest claims machine-proved. This note opens the wider front. It states the reading that motivates the whole program, fixes the epistemic ground rules under which that reading may appear in scientific work, maps the places where K and S already touch physics through established results, and lays out candidate experiments, in Idris2 and on the existing engine, whose outcomes would be findings in their own right. The strategic model is Leibniz: the binary arithmetic he read theologically in 1697 persuaded the world on its mathematics alone. The program's claim is not that the calculus proves the reading. The claim is that the reading is generative, and that its fruits can stand before any scientist without it.

1. The reading, stated once

The reading we bring to this work is the following. Before form, there is a union that encompasses everything. Separation enters as an idea: the move that takes two things and discards one, keeping the other apart from the whole. The answer to separation is not the denial of form but its reconciliation: the move that takes one thing and lets it flow undivided into every branch that needs it, so that unity persists inside multiplicity. In the calculus these two moves have names. K x y reduces to x: it is pure separation, the discarding of y. S x y z reduces to x z (y z): it is pure distribution, one z given whole to both futures at once. Everything computable is built from these two moves and nothing else. That is the sense in which we take {K, S} to be as close to a world formula as a formal system has come.

Three facts already inside the project give this reading its first anchors, and all three are machine-checked or classical.

First, identity is not primitive. The identity combinator is I = S K K, and the project's proof module verifies skIdentity: S applied to two separations restores the thing itself, unchanged. In the reading: unity is not an axiom of the system; it is recovered, exactly, as the answer of S to K.

Second, the two moves are the two axioms of implication. Under the Curry-Howard correspondence, K inhabits A implies (B implies A), and S inhabits (A implies (B implies C)) implies ((A implies B) implies (A implies C)); these are precisely the two axioms with which Frege opened the Begriffsschrift in 1879, forty-five years before Schoenfinkel isolated the combinators [3, 6]. The logic of 'if' is the logic of these two moves, discovered twice, independently, from opposite directions.

Third, the dichotomy reappears at the arithmetic level. In R_arith of paper 1, the rule for multiplication by a successor duplicates the multiplicand (R1: v occurs twice on the right), the rule for multiplication by zero erases it (R2: v occurs zero times), and both addition rules are linear (R3, R4: every parameter occurs exactly once) [1]. Multiplication carries the S-face and the K-face of the calculus into number itself; addition touches neither. This is a checkable statement about occurrence counts in rules that are already compiler-verified.

Nothing below proves the reading. What can be demonstrated is that the reading is generative: that taking separation and distribution as the primitive vocabulary leads to theorems, measurements,

and structures that stand on their own. Leibniz is the model and the warning at once. He read the binary system as creation from nothing, struck a medal for it, and sent it to the Jesuits in China; what survived and conquered was the arithmetic [7]. We aim for the same asymmetry, deliberately.

2. Epistemic ground rules

The program keeps two registers and never mixes them. Papers speak mathematics and physics; this note holds the reading. A result enters a paper when it is a theorem, a machine-verified computation with stated bounds, or a precise correspondence with at least one consequence that was not visible before. Motivation belongs to the context of discovery; the papers live in the context of justification. This is not concealment. It is the condition under which the work can be judged, and the only route to the authority the program seeks.

Any scientist worth persuading will raise three objections, and the program answers them in advance rather than around them.

The base is not unique. Combinatory completeness is shared by {B, C, K, W}, by {S, K, I}, and even by single combinators such as the one-point bases built from iota, where iota x = x S K. Two honest replies. The single-combinator bases are encodings that factor through S and K conceptually and pay for their unity in size; and among the natural bases, {K, S} is the minimal one whose two elements have orthogonal semantic roles, deletion and distribution, which is exactly the axis the program studies. The convergence of Frege's axioms, Schoenfinkel's combinators, and Curry's systems on the same pair is historical evidence of canonicity, not proof. Uniqueness claims will be argued where needed and asserted nowhere.

Universality deflates. Since everything Turing-complete simulates everything else with constant overhead, the invariance theorem of algorithmic information theory says the choice of substrate changes descriptions only by O(1) [16]. Reply: the program's claims are not about what can be computed but about which structures become visible under this decomposition. The erasure and duplication accounting of Section 4 is well-defined precisely because the base fixes what counts as an erasure. Generativity, not exclusivity, is the testable property.

Grand rewriting claims have a reception problem. The nearest living relative of 'rewriting as world formula' is the Wolfram program, hypergraph rewriting offered as fundamental physics [18], and its reception teaches the lesson this note already adopted: concrete lemmas travel, cosmologies do not. The program leads with findings of bounded scope and lets the frame stay in this note.

3. Where K and S already touch physics

The contact points below are established science. The reading did not create them; it selects and organizes them, and each one suggests measurements the existing engine can make.

3.1 K and the arrow of time. Landauer's principle says erasing one bit dissipates at least kT ln 2 of heat [12]; Bennett showed computation can be reversible exactly insofar as it avoids erasure [13]. In the calculus, K is the erasing move: every K-firing discards a subterm, and every S-firing duplicates one. The affine fragment (no duplication) and the linear fragment (neither erasure nor duplication) are the combinatory homes of reversibility-friendly computation, and they are classical objects: BCK and BCI logic, both treated in Bimbó [5]. In the reading: the thermodynamic arrow enters computation exactly where separation does. What is measurable: the erasure and duplication profile of any reduction, term by term, exactly.

3.2 Duplication, erasure, and the quantum boundary. Quantum information forbids cloning unknown states and, dually, deleting them; the linear fragment is where quantum processes live, and the categorical semantics of quantum protocols is built on exactly this restriction [8, 9, 10, 11]. In combinatory terms: the quantum world is the world where neither K's discard nor S's free duplica-

tion acts on unknown data, and classicality arrives precisely with copyability. This is the sharpest established sentence the program owns: the classical-quantum boundary, in the resource-theoretic reading, is the availability boundary of K and of S-duplication. A precision piece stating this correspondence carefully, with the project's boxes and probes classified by fragment, would be synthesis rather than discovery, but synthesis of citable clarity.

3.3 Rewriting as substrate. Zuse's Rechnender Raum and the digital physics line [17], and the Wolfram hypergraph program [18], are the standing attempts to make rewriting fundamental; confluence appears there under the name causal invariance. The project relates to this line the way a lemma relates to a manifesto, and intends to keep it that way. What transfers inward is one good question: which physical symmetries have analogues as confluence statements of small rewriting systems.

3.4 Algorithmic chemistry. Fontana and Buss built a proto-biology from lambda terms colliding in a well-mixed reactor and found self-maintaining organizations [14]; Kruszewski and Mikolov ran the same idea directly on S and K and observed emergent autocatalytic structures [15]. This is the direction closest to new findings with existing tooling: the project already owns a fast reducer, a term enumerator, and orbit-signature deduplication from the E4 campaign [2]. A combinator soup is a census with dynamics.

3.5 Program-size and priors. Tromp's binary encodings make {S, K} a concrete universal description language with exact sizes [16]; algorithmic probability turns any such language into a prior over structures. The project's contribution here is honesty about invariance (Section 2) plus exact size accounting for its own arithmetic representations, unary against binary, continuing paper 2' s cost story.

4. Candidate experiments

All experiments are Idris2, per project standard, and reuse the engine of papers 1 and 2.

E6, the combinator soup. Delivered as the program's second results paper [29]: three reactor runs, monoculture collapse in all three, two machine-verified self-reproducing absorbers, and the selection mechanism named; E6.2, the conserving reactor, is delivered in the same paper's second half [29]: balanced reactions with compiled conservation theorems, two frozen regimes diagnosed on the way, a driven steady state, the absorbers extinct under the resource tax, and the identity family persisting as the standing crop. Both protocols ran as drafted. A multiset of terms; a step draws two, applies one to the other, reduces with bounded fuel, and returns the results. Instruments: population statistics over time, motif recurrence through the orbit signatures of E4, detection of self-maintaining sets in Fontana's sense, and autocatalysis tests (does the presence of t raise the production rate of t). Deliverable: census-style tables of emergent structures, with every reduction certified by the existing discipline. Risk: moderate; the phenomenon is documented in the literature [14, 15], so the finding would be its precise anatomy under this engine, including whether identity-like and pairing-like forms dominate the persistent core.

E7, the entropy budget of arithmetic. Instrument the reducer to count K-firings and S-firings, weighting each K-firing by the size of the discarded subterm. Run the arithmetic of both papers, unary numerals against binary numerals, and chart erasure and duplication against n. Landauer prices the erasures; the counting itself is exact and machine-checked, the joules are a reading. Conjectured shape: representation choice moves cost between duplication and erasure rather than removing it, sharpening paper 2' s encoding-is-not-neutral theme into a thermodynamic statement.

E8, the linearity ladder. Delivered as the program's first results paper [19]; the plan below stands as drafted and executed. Classify every combinator and box of the project by fragment membership:

linear (BCI), affine (BCK), full {S, K}. Formal core, small and checkable: an occurrence-count lemma over the rule set, verifying the Section 1 observation that R1 duplicates, R2 erases, and R3 and R4 are linear. Extension: which parts of the arithmetic survive in the affine fragment, and what breaks first without K. This experiment is nearly free and produces the cleanest sentence the program has: in this arithmetic, multiplication is where separation and distribution enter number.

E9, the mathematics that compels. The standing Tier 3 of the formalization plan: strong normalization of R_arith through the skeleton homomorphism and a recursive path ordering, Church-Rosser for weak reduction, and the fundamental theorem of arithmetic module that makes Proposition 4 unconditional. These are the results a rewriting theorist recognizes without any frame at all, and they remain the shortest road to authority.

5. Priorities

E8 is delivered [19]: the compiler-checked split of the rule set, the exact non-linear budget of multiplication, and the program's best sentence are in print. E6 and E6.2 are delivered [29]: the collapse understood, the remedy built, the absorbers extinct under conservation, and the driven vessel's standing crop is the identity family. The null-model census of the driven steady state joins the queue; E7 next, feeding on both reactors' instrumentation. E9 runs in parallel as the mathematical backbone. The precision piece of 3.2 waits until E8 gives it the project-internal classification it needs.

6. Two rows of publication

The program will publish in two rows. Row A is the working papers: mathematics and, later, physics-adjacent findings, written for scientists, held to the standards of Section 2, and never citing Row B. Row B is interpretation for everyone: the class of book that physics has produced for a century and that mathematics and computer science almost never have. Physics readers can choose among Eddington and Jeans, who wrote as idealists at the height of their scientific authority [20, 21], Capra's reading of quantum theory through Eastern thought [22], and Davies' theological questions asked from inside cosmology [23]. The computing shelf has Hofstadter's proof that deep formal material can carry a book of ideas for a general reader [24], and little else in this register. The gap is real, and a calculus of two moves with everything built from them is unusually well shaped to fill it.

Row B's position, stated plainly. The current default worldview treats matter and energy as primary and everything else as derivative. That view is not a result of science; it is an interpretation laid over the mathematics, and it faces standing philosophical difficulties: the concept of matter has no stable definition to be primary about (Hempel's dilemma [25]), and experience itself resists derivation from it (the hard problem [26]). Row B offers the opposite interpretation, in harmony with the spiritual teaching of *A Course in Miracles* and *A Course of Love* [27, 28]: mind first, separation as the one idea that seems to shatter unity, and the Answer already present inside form. The calculus supplies the exact vocabulary. K is the move of separation. S is the move that gives one thing whole to many futures. Identity is not primitive but restored, I = S K K, and the arithmetic of the papers shows precisely where the two moves enter number [19].

Rules for Row B, extending Section 2. The books never claim that the mathematics proves the teaching; they claim that it reads, exactly and beautifully, and that the materialist alternative is also only a reading. Capra's reception is the standing lesson: the moment 'resonates with' hardens into 'confirms', scientists stop listening and are right to. The books state the materialist position at its strongest before answering it; a reader must recognize their own view in it before it is questioned. The books are pro-science without reservation: every formal claim carries a pointer into Row A, and nothing in Row A ever depends on Row B. And Row B publishes behind Row A: the precedent that

matters is that Eddington wrote with his scientific authority already banked. Drafting begins now; publication follows the program's first recognized results.

Book one, working sketch. Title, decided: *Rise of Form*, with Two Moves as the opening chapter; the earlier candidates are retired. Arc: the two moves introduced humanly, discard and share, with the double discovery by Frege and Schoenfinkel (Chapter 1); what a world formula can honestly mean, and materialism as one interpretation among two (Chapter 2); separation, erasure, and the arrow of time, with Landauer's price on every K firing (Chapter 3); the Answer, sharing, and identity restored (Chapter 4); number, with primality as unreachability and the ladder finding that multiplication carries both faces while addition touches neither [19] (Chapter 5); the quantum boundary as the place where copying and discarding are refused (Chapter 6); the combinator soup and the absorbers that devour it, now that E6 has reported [29] (Chapter 7); the reading itself, stated openly at last, as invitation and not proof (Chapter 8).

7. Relation to papers 1 and 2

Papers 1 and 2 remain what they are: mathematics with machine-checked cores and no metaphysics [1, 2]. This note is the program's private compass and public strategy in one document: it says why the work is done, and it commits the work to standards under which the why is never load-bearing. If the reading is true in any sense that matters, the calculus will keep producing structure worth publishing; if the structure stops coming, the reading loses its claim on working time. That is the falsifiability a research program can honestly offer, and it is the one this note accepts.

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